Replicating Jones Models

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1. Transects were filtered to only include broadcast burning, wildfire, or Untreated points.

Note: Incomplete transects (those with thinning-only points filtered out) were allowed in the dataset. EVT groups without broadcast/wildfire treatments were removed from controls. The EVT group "other" was also removed for consistent context.

2. Six models were constructed for different combinations of starting severity and transect Type:

Low x Entering, Moderate x Entering, High x Entering Low x Exiting, Moderate x Exiting, High x Exiting

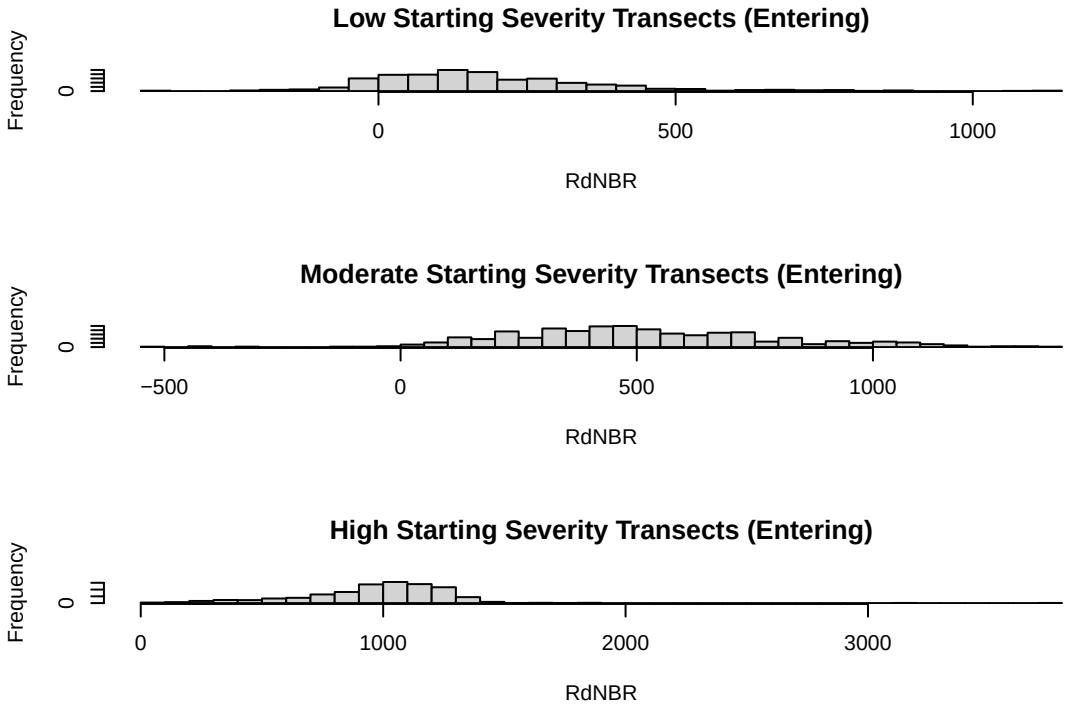
3. The model formula is:

RdNBR ~ Type + Treated + (1|Incid_Name/Transect_ID)

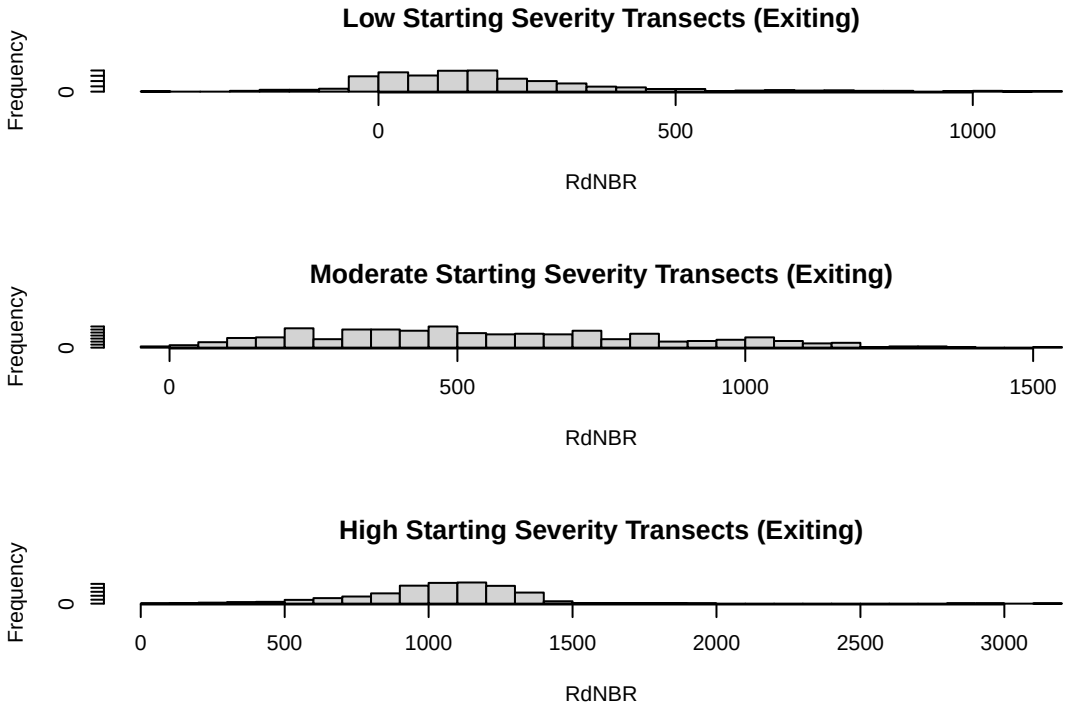
Where Type = control or entering/exiting & Treated = Untreated or broadcast/wildfire

Jones Dataset

Entering Datasets



Exiting Datasets



Entering Models

Low Entering Model

RdNBR

Predictors

Estimates

CI

p

(Intercept)

198.13

138.03 – 258.24

<0.001

Type [entering]

-24.37

-74.74 – 26.00

0.343

Treated [Yes]

-2.13

-40.18 – 35.92

0.913

Random Effects

2

22316.61

00 Transect_ID:Incid_Name

9077.79

00 Incid_Name

5270.85

ICC

0.39

N Transect_ID

82

N Incid_Name

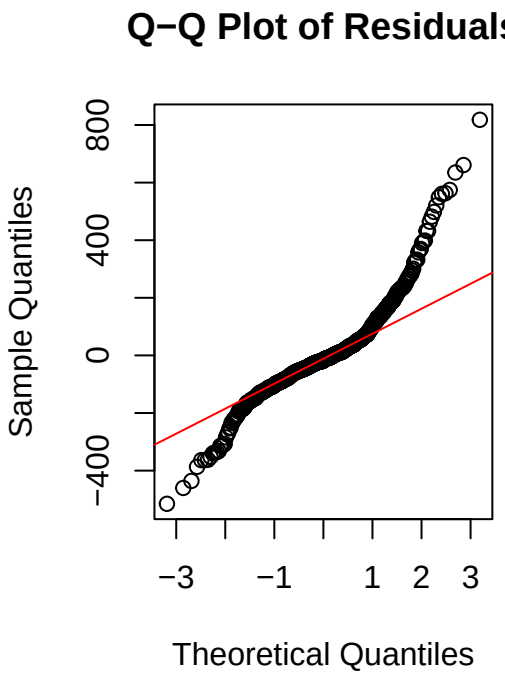
11

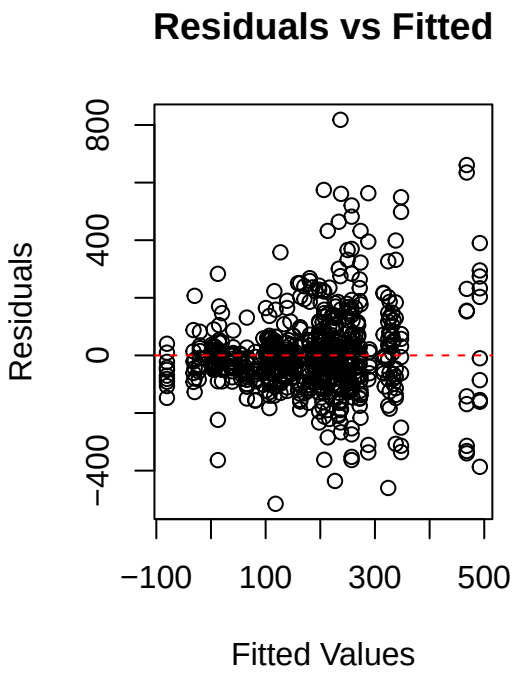
Observations

700

Marginal R2 / Conditional R2

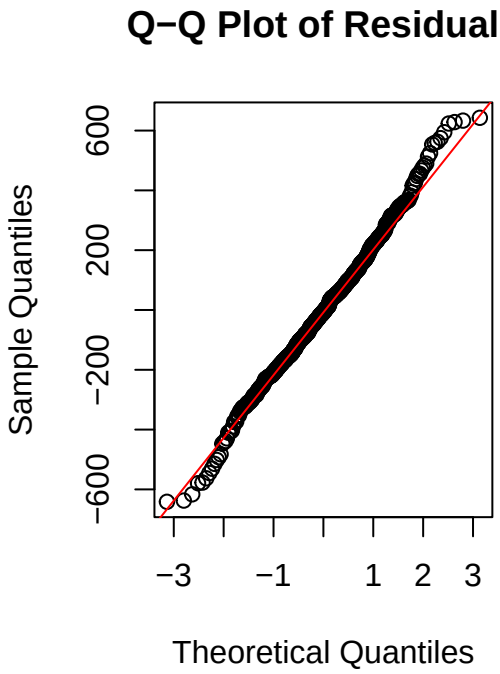
0.004 / 0.394

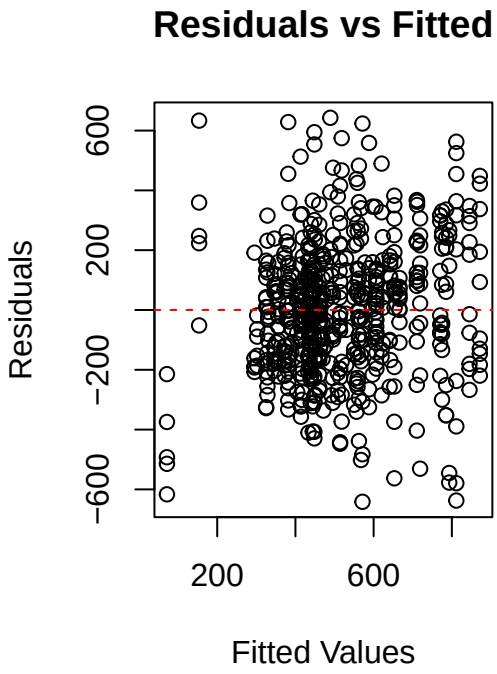




Moderate Entering Model

RdNBR
Predictors
Estimates
CI
p
(Intercept)
558.11
491.87 – 624.35
<0.001
Type [entering]
-63.00
-147.07 – 21.07
0.142
Treated [Yes]
-82.21
-148.70 – -15.72
0.015
Random Effects
2
53556.28
00 Transect_ID:Incid_Name
23037.83
00 Incid_Name
954.43
ICC
0.31
N Transect_ID
71
N Incid_Name
10
Observations
585
Marginal R2 / Conditional R2
0.032 / 0.332





High Entering Model

RdNBR			
Predictors	Estimates	CI	p
(Intercept)	967.42	878.57 – 1056.26	<0.001
Type [entering]	-22.65	-87.96 – 42.66	0.496
Treated [Yes]	-357.87	-415.96 – -299.77	<0.001

Random Effects

2

53136.90

00 Transect_ID:Incid_Name

21208.89

00 Incid_Name

5717.72

ICC

0.34

N Transect_ID

110

N Incid_Name

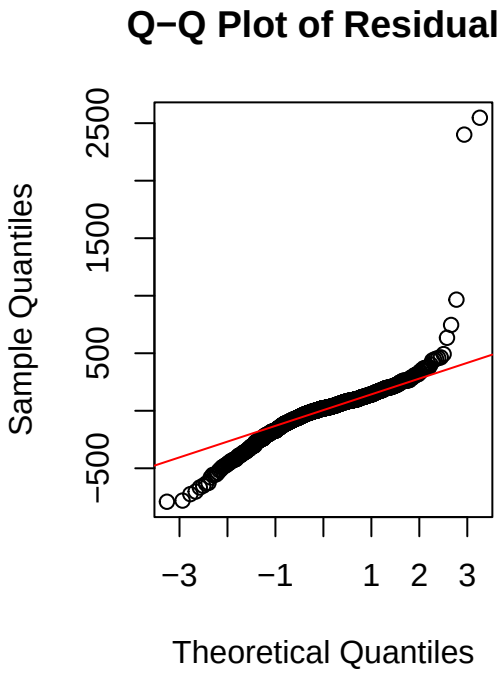
7

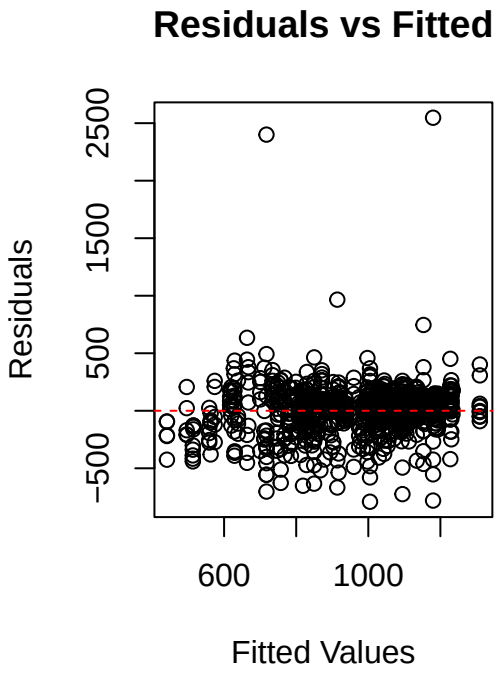
Observations

903

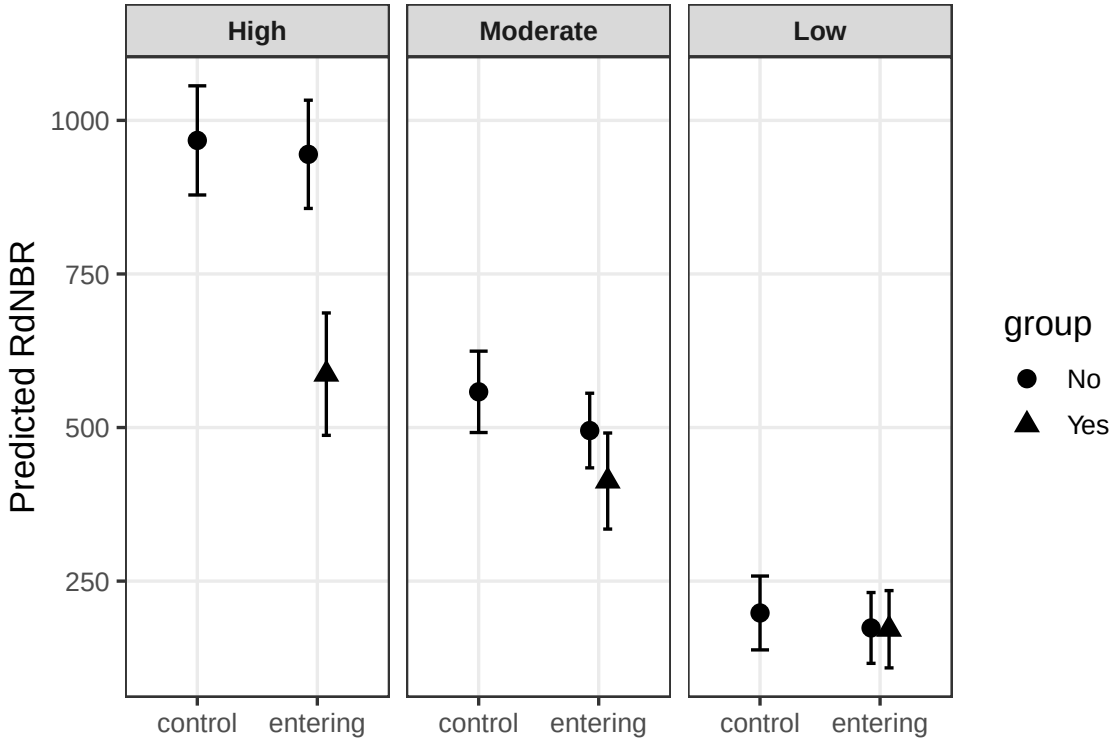
Marginal R2 / Conditional R2

0.155 / 0.439





Entering Effect



For GSS Abstract, percent drop in severity when high severity fire enters a Treated area

```
##          x group predicted conf.low conf.high
## 1  control    No  967.4159 878.5670 1056.2648
## 2  control   Yes  609.5501 502.4961  716.6041
## 3 entering    No  944.7644 856.6611 1032.8677
## 4 entering   Yes  586.8986 487.2670  686.5301
```

```
##  abs_drop  pct_drop
## 357.86584  37.87884
```

Exiting Models

Low Exiting Model

RdNBR

Predictors

Estimates

CI

p

(Intercept)

185.59

107.56 – 263.63

<0.001

Type [exiting]

-20.47

-93.58 – 52.64

0.583

Treated [Yes]

-24.01

-75.67 – 27.65

0.362

Random Effects

2

25521.72

00 Transect_ID:Incid_Name

17563.56

00 Incid_Name

8104.29

ICC

0.50

N Transect_ID

68

N Incid_Name

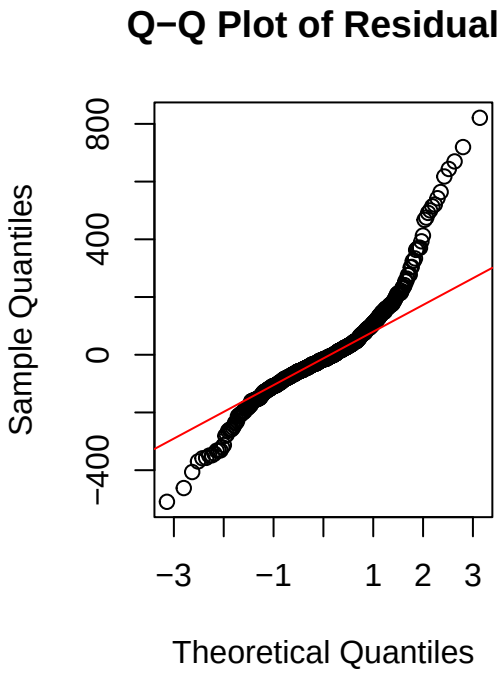
11

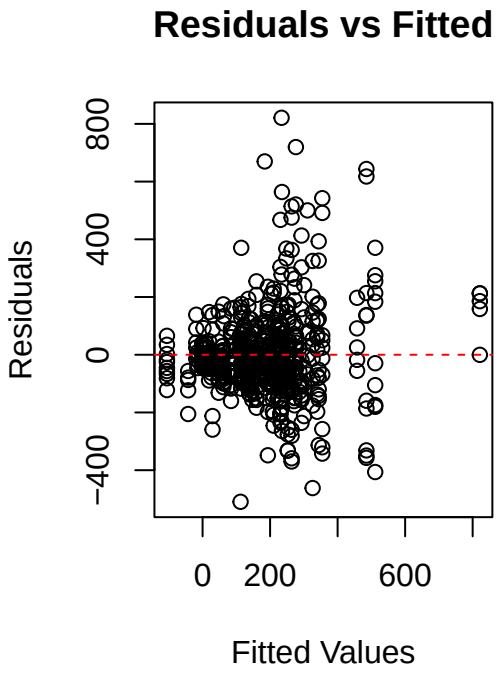
Observations

590

Marginal R2 / Conditional R2

0.005 / 0.504





Moderate Exiting Model

RdNBR

Predictors

Estimates

CI

p

(Intercept)

559.95

480.97 – 638.93

<0.001

Type [exiting]

60.45

-59.52 – 180.41

0.323

Treated [Yes]

-214.11

-303.33 – -124.88

<0.001

Random Effects

2

52760.98

00 Transect_ID:Incid_Name

40420.62

00 Incid_Name

221.25

ICC

0.44

N Transect_ID

53

N Incid_Name

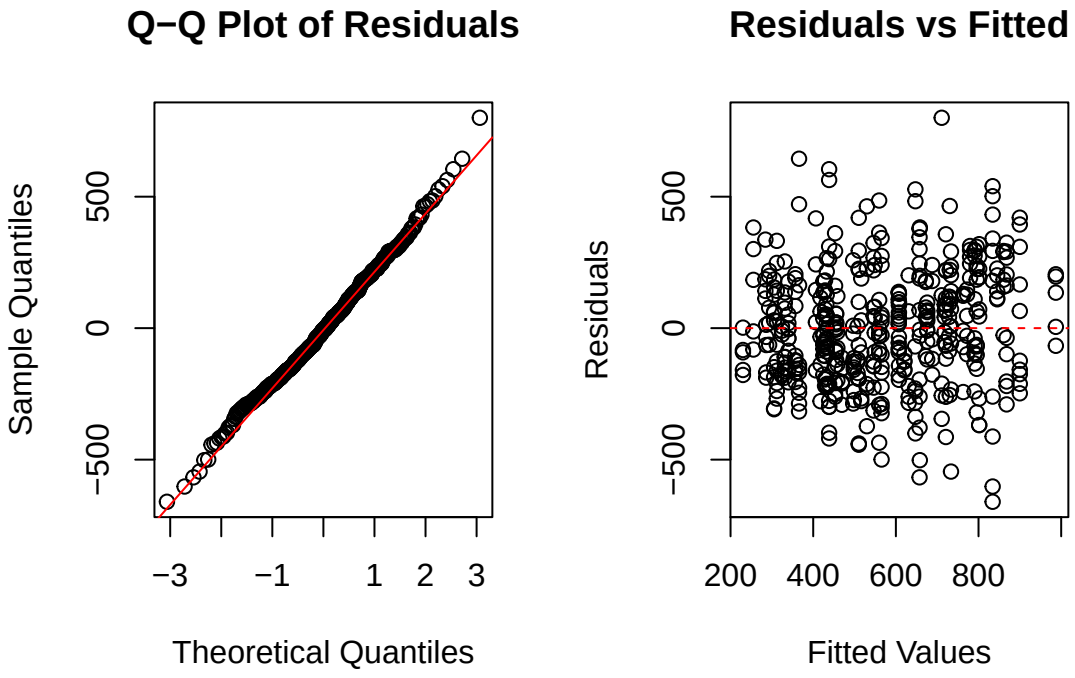
8

Observations

457

Marginal R2 / Conditional R2

0.035 / 0.455



High Exiting Model

RdNBR

Predictors

Estimates

CI

p

(Intercept)

923.46

765.20 – 1081.71

<0.001

Type [exiting]

82.89

-29.96 – 195.74

0.150

Treated [Yes]

-249.30

-423.57 – -75.03

0.005

Random Effects

2

42490.20

00 Transect_ID:Incid_Name

44971.04

00 Incid_Name

21290.27

ICC

0.61

N Transect_ID

76

N Incid_Name

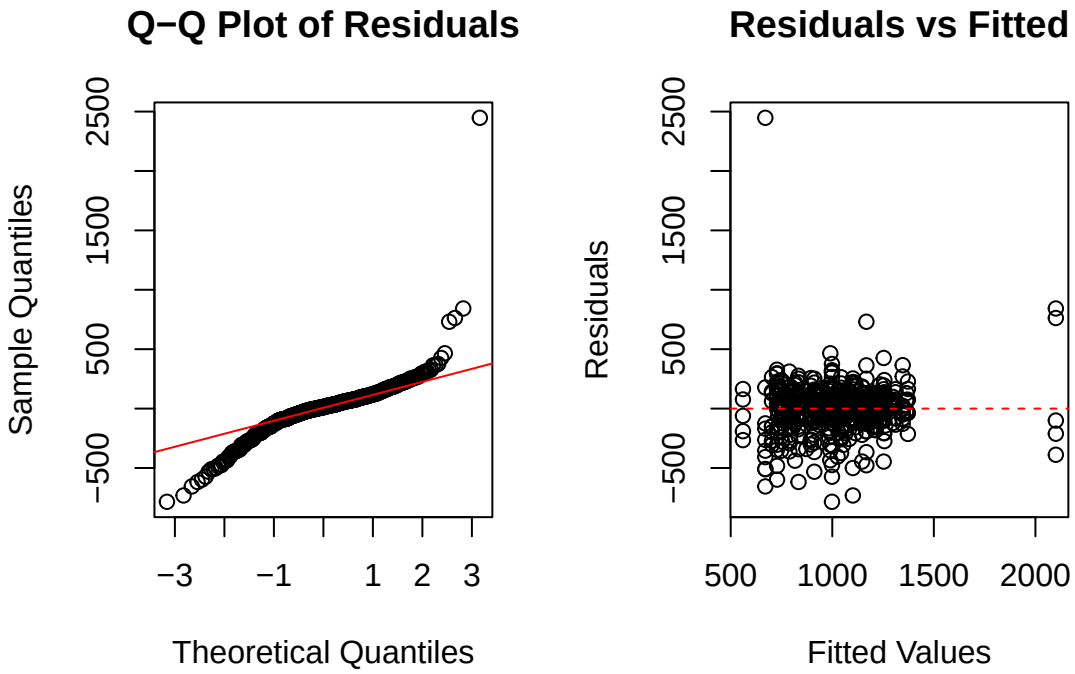
6

Observations

635

Marginal R2 / Conditional R2

0.015 / 0.615



Exiting Effect

